

HR Mail Intelligence Agent

■ 1. Problem Overview

As a final-year engineering student actively pursuing internship and placement opportunities, staying on top of important emails from HR teams and companies is critical — yet extremely time-consuming. Inboxes receive dozens of messages daily, and manually monitoring them for job-related communications leads to missed opportunities and delayed responses.

The core problem is the **absence of a real-time, intelligent filtering mechanism** that can distinguish high-priority HR and recruitment emails from general inbox noise and deliver instant notifications to the candidate.

■ 2. Problem Statement

"Design and implement an intelligent email monitoring agent using n8n that continuously watches a Gmail inbox, automatically identifies emails from HR professionals, recruiters, or companies — especially those related to mass hiring, campus placements, internships, or job opportunities — and delivers real-time structured alerts to the user via a Telegram bot."

■ 3. Context & Motivation

- Students receive placement-related emails at irregular hours; manual monitoring is inefficient.
- HR emails often contain deadlines, interview links, and offer details requiring prompt action.
- Mass hiring drives by large companies arrive in bulk and can easily be lost among promotional mail.
- A Telegram notification is persistent, accessible on mobile, and does not require opening a browser.
- n8n provides a low-code automation backbone with powerful conditional logic and API integrations.

■ 5. Proposed n8n Workflow Architecture

The solution is implemented as an n8n automated workflow consisting of the following nodes:

1	Schedule Trigger Fires every 5–10 minutes to initiate the workflow.
2	Gmail Node (Read Emails) Fetches unread emails from the inbox using OAuth2. Filters for emails received since last run.
3	IF / Switch Node Checks sender domain, subject keywords, and email labels to route HR/company emails separately.
4	AI / Text Classifier Node Optional LLM-powered step to analyze email body for job-related intent (mass hiring, offer letters, interview calls).
5	Function Node (Formatter) Extracts and formats: Sender Name & Email, Subject Line, Short Snippet (first 200 chars), Timestamp, and Gmail Link.
6	Telegram Node (Send Message) Sends a rich-text Telegram message to the user's bot with emoji-formatted alert layout.
7	Google Sheets / Database Node Logs processed email IDs to prevent duplicate notifications across runs.

